WiFi Shuriken

A New Architecture For High Performance Scanning

CoD_Segfault

Contents

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Who Am I?

CoD_Segfault

- WiGLE Rank: 41 (as of 2026-07-27)
- Author of BW16-Open-AT
- Code Contributor to Wardriver.uk
- Hardware Designer
  - Mini Wardriver v1 and v2
  - WiFi Shuriken
- Hard Hat Brigade Co-Organizer

02

What is the Wardriving?

What is Wardriving?

- Observation of Wireless Devices
  - WiFi Access Points
  - Bluetooth Devices
  - Cellular Networks
- Metadata Collection
  - SSID & BSSID
  - Band and Channel
  - Security Set
  - Signal Strength
  - Location
  - Date and Time

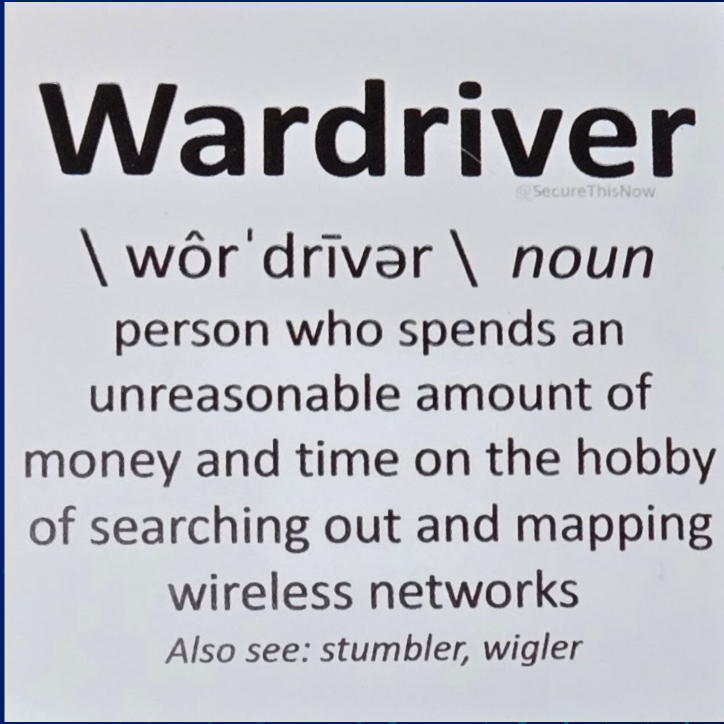


Image Credit: MrBill

What To Do With the Data?

- WiGLE – Wireless Geographic Logging Engine
  - User-Submitted Submissions
  - Catalogs and Indexes Observed Networks
  - Trilaterates to Determine Location
  - Displays Results on a Map
  - Generates Stats with Trends Over Time

Why?

- Fake Internet Points!
- Collectively See Trends
- Explore New Areas
- It’s Fun

¯_(ツ)_/¯

03

What is the WiFi Shuriken?

Where Did it Come From?

- Wardriver.uk by Joseph Hewitt
- WiFydra by Lozaning

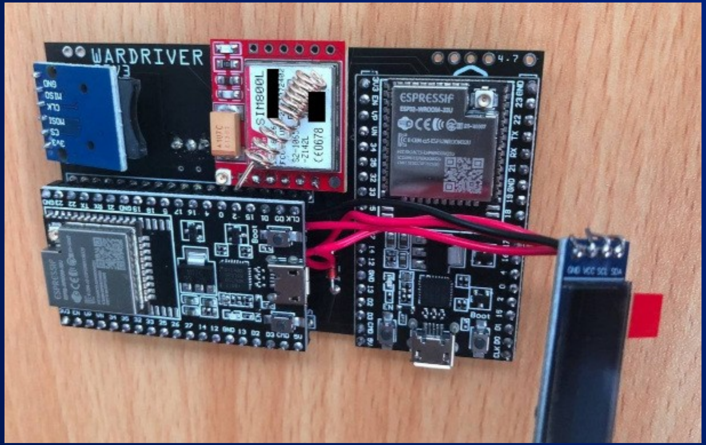


Image Credit: Joseph Hewitt

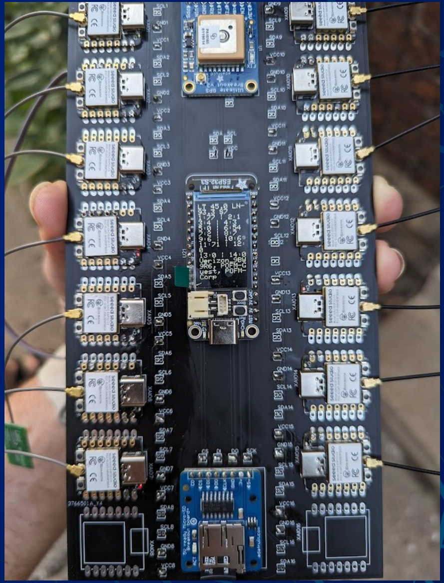
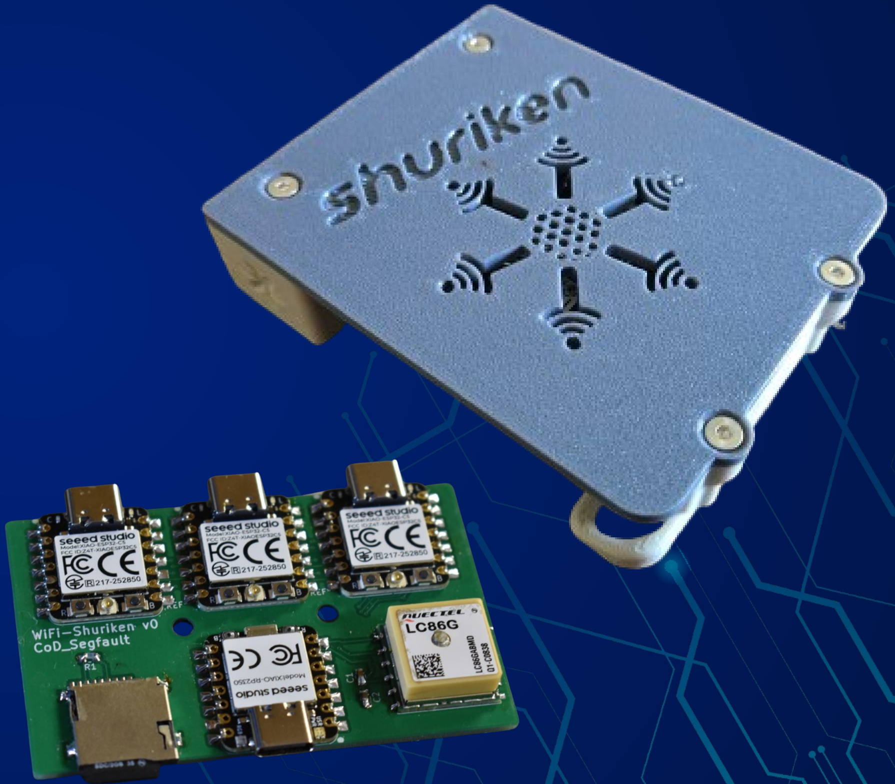


Image Credit: Lozaning

What is the WiFi Shuriken?

- Microcontroller-based Wardriving Platform
- Standalone Operation
  - Six Dual Band WiFi Radios
  - Multi-Constellation GNSS
  - SD Logging
- Controller and Scanner Architecture
- Engineered to Remove Bottlenecks



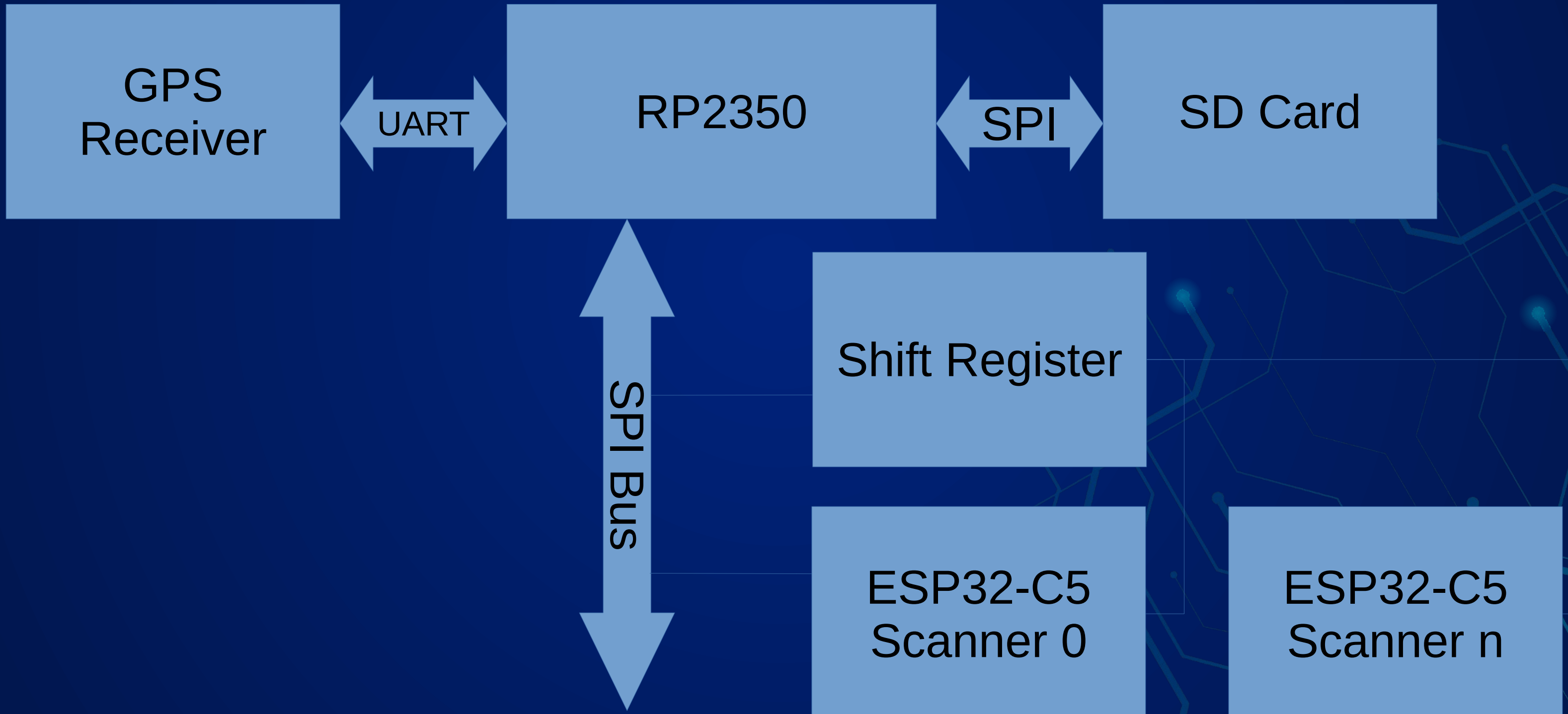
What Makes the Shuriken Unique?

- Extreme Separation of Responsibilities
- Intentionally Limited Scope
- Fast, Direct Communication
- Fault-Tolerant
- Scalable
- Future-Proof

Operation

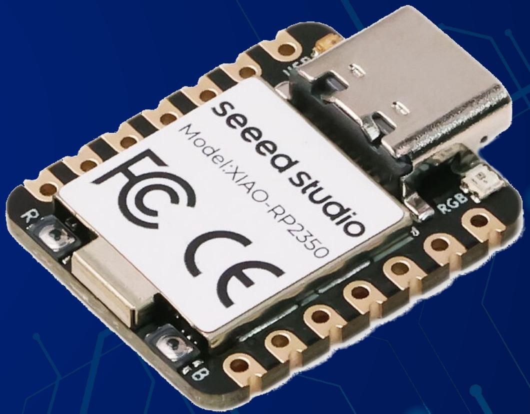
- Apply Power; Scanning and Logging Begins
- Deduplication of Observed Devices
  - SSID – Network Name
  - BSSID – MAC Address
  - Band and Channel
  - Security Set
- Fault Toleranant
  - No Single Scanner is Responsible for Single Channel

Shuriken Architecture



Shuriken Controller

- Seeed Studios XIAO RP2350 (Raspberry Pi Pico 2)
  - Dual-Core ARM Processor
  - Used in DC32 Badge
- Core 0
  - SD Card Logging
  - GPS Location, Time, and Date
  - Optional Phone/PC Integration (In Development)
- Core 1
  - Scan Channel Scheduler
  - Scanner Communication

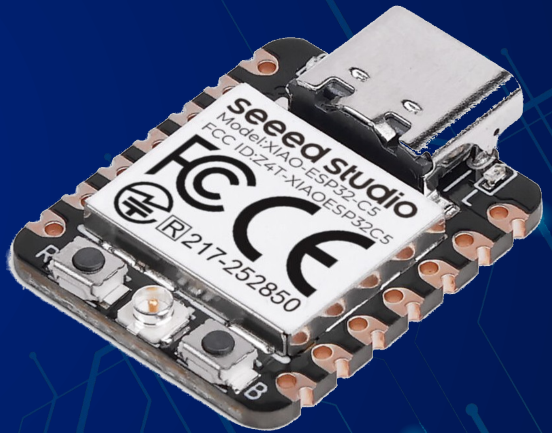


GNSS Receiver

- Quectel LC86L
- Multi-Constellation
  - GPS, QZSS, BeiDou, GLONASS, Galileo
  - 3 Simultaneous Constellations
- Built-in RTC
- Battery-backed Memory
- Configuration Command Set
  - 115200 Baud Rate
  - 5Hz Updates
  - Non-essential Messages Disabled

Shuriken Scanners

- Seeed Studios XIAO ESP32-C5
    - Single-Core RISC-V Processor
    - Dual-Band WiFi Radio
- Accepts Commands From Controller
- Handles All WiFi Scan Logic
    - Radio Control (Active/Passive, Scan Duration)
    - Queues Results Until Queried



What the Shuriken Doesn’t Have

- LCD/OLED Display
- Web Interface/Auto Upload
- Runtime Config

The Secret Sauce – SPI Bus and Protocol

- SPI Bus Runs at 20MHz
  - Bi-directional
  - 20Mbps (~2.5 MB/sec)
  - At 70% Utilization: 27,000 Results/sec
- Controller Talks Directly to Scanners
  - Shift Register Addresses Individual Scanner (Daisy-chainable to 64+)
  - Scanners Only Transfer Data When Prompted
- Simple Protocol
  - Results Count/Status
  - Transfer Record
  - Scan Channel

The Secret Sauce – Deduplication

- Multi-tier Deduplication
  - Each Scanner Only Reports AP once
  - Controller Deduplicates Results Before Logging
- Hash Table with FIFO eviction
  - Hashing Shrinks Memory Footprint
  - Speeds Up Lookups

Scalable Future

- Potential Future Scanners
  - HaLow
  - 6 GHz
  - ???
- More Scanners
  - Trivial to Add 64+ Scanners

04

How Does it Compare?

Samsung S24+ Comparison

Metric	S24+	Shuriken	Comparison
Handheld College Campus 2.4 GHz	384	578	+194 (+50.50%)
Handheld College Campus 5 GHz	344	532	+188 (+54.70%)
Handheld College Campus Combined	728	1,110	+382 (+52.50%)
Car 2.4 GHz	8,393	14,209	+5816 (+69.30%)
Car 5 GHz	13,454	15,825	+2371 (+17.60%)
Car Combined	22,507	30,032	+7525 (+33.40%)
Downtown 2.4 GHz	14,749	22,956	+8,207 (+55.6%)
Downtown 5 GHz	36,762	44,786	+8,024 (+21.8%)
Downtown Combined	51,511	67,724	+16,213 (+31.5%)
Downtown (1 minute highest) 2.4 GHz	215	606	+391 (+181.9%)
Downtown (1 minute highest) 5 GHz	846	1,442	+596 (+70.4%)
Downtown (1 minute highest) Combined	1,061	2,048	+987 (+93.0%)

Wardriver.uk Comparison

Metric	Wardriver.uk	Shuriken	Comparison
Suburban 2.4 GHz	6,314	5,910	-404 (-6.4%)
Suburban 5 GHz	5,218	6,894	+1,676 (+32.1%)
Suburban Combined	11,532	12,804	+1,272 (+11.0%)
Downtown 2.4 GHz	17,459	9,685	−7,774 (-44.5%)
Downtown 5 GHz	4,448	21,197	+16,749 (+376.6%)
Downtown Combined	21,907	30,879	+8,972 (+41.0%)

Kismet Comparison*

Metric	Kismet	Shuriken	Comparison
Morning Run #1 – 2.4 GHz	11,035	12,108	+1,073 (+9.7%)
Morning Run #1 – 5 GHz	15,052	24,760	+9,708 (+64.5%)
Morning Run #1 – Total	26,087	36,868	+10,781 (+41.3%)
Morning Run #2 – 2.4 GHz	10,799	12,375	+1,576 (+14.6%)
Morning Run #2 – 5 GHz	15,059	22,691	+7,632 (+50.7%)
Morning Run #2 – Total	25,858	35,066	+9,208 (+35.6%)
Evening Run #1 – 2.4 GHz	14,891	16,784	+1,893 (+12.7%)
Evening Run #1 – 5 GHz	18,294	33,786	+15,492 (+84.7%)
Evening Run #1 – Total	33,185	50,570	+17,385 (+52.4%)
Evening Run #2 – 2.4 GHz	14,891	18,007	+3,116 (+20.9%)
Evening Run #2 – 5 GHz	18,294	33,640	+15,346 (+83.9%)
Evening Run #2 – Total	33,185	51,647	+18,462 (+55.6%)

WWWD Results

Device	2.4 GHz	5 GHz	6 GHz	All bands
Shuriken AC1177	27,870 (324)	34,639 (720)	—	62,498 (1,044)
Shuriken 41ECED	27,759 (319)	34,543 (917)	—	62,287 (1,236)
Galaxy S24+	19,964 (124)	32,922 (1,082)	2,054 (2,054)	54,938 (3,259)
wardriver.uk	29,113 (1,013)	17,694 (60)	—	46,806 (1,073)
All-device union	30,299	37,524	2,054	69,857

Thank You Testers!

- grim0us
- M0nkeyDrag0n
- MrBill
- pejacoby

SURPRISE!

- Upgrade your existing wardriver.uk units with an ESP32-C5!
- C5-Open-AT replaces the Realtek BW16 for 5GHz scanning!
- Available NOW at :

  https://github.com/CoD-Segfault/C5-Open-AT

```
Welcome to C5-Open-AT!
ATWS
AP : 1,SouthwestWiFi,36,Open,-80,68:28:CF:16:9B:70
AP : 2,2Wire43613,36,WPA2 Enterprise,-80,68:28:CF:16:9B:71
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[ATWS]
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Q&A

Thanks!

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